

Weekly meeting progress

Friday, Aug 14, 2026





What did I do this week?

- Fixed RPC communication between the router and the agent
- Optimized training time:
 - Training with 2, 3, or 4 replicas was 3 times slower than with 5 or 6 replicas
 - This was due to the index selection under the storage budget limit (improved knapsack solver)
- Ran the project multiple times on CloudLab to validate performance



Benchmarked the classical version of the algorithm on a 10 GB TPC-H database with a 5 GB index budget, with 1,100 queries, using budget mode ignore during training

Total queries runtime in seconds on PostgreSQL use the same benchmark script qDINA project:

	2	3	4	5	6
Exp 1	3176	3401	2125	1602	1927
Exp 2	3226	2265	2559	1497	1738
Mean	3201	2833	2342	1549.5	1832.5
DINA	3900	3700	3000	3500	3400



Algorithm names

DINA-S : **D**ivergent **I**ndex **A**dvisor using Deep Reinforcement Learning with
Routing



Next tasks

- Run the project several times to get reliable performance results
- Test other storage budgets (for example, 15% and 25% of the database size) to see how they affect the results
- Keep writing the final report for the internship.